IS3 - 139

RAIR-VF: Reasoning-Augmented Interactive Image Retrieval with Visual Fact Grounding

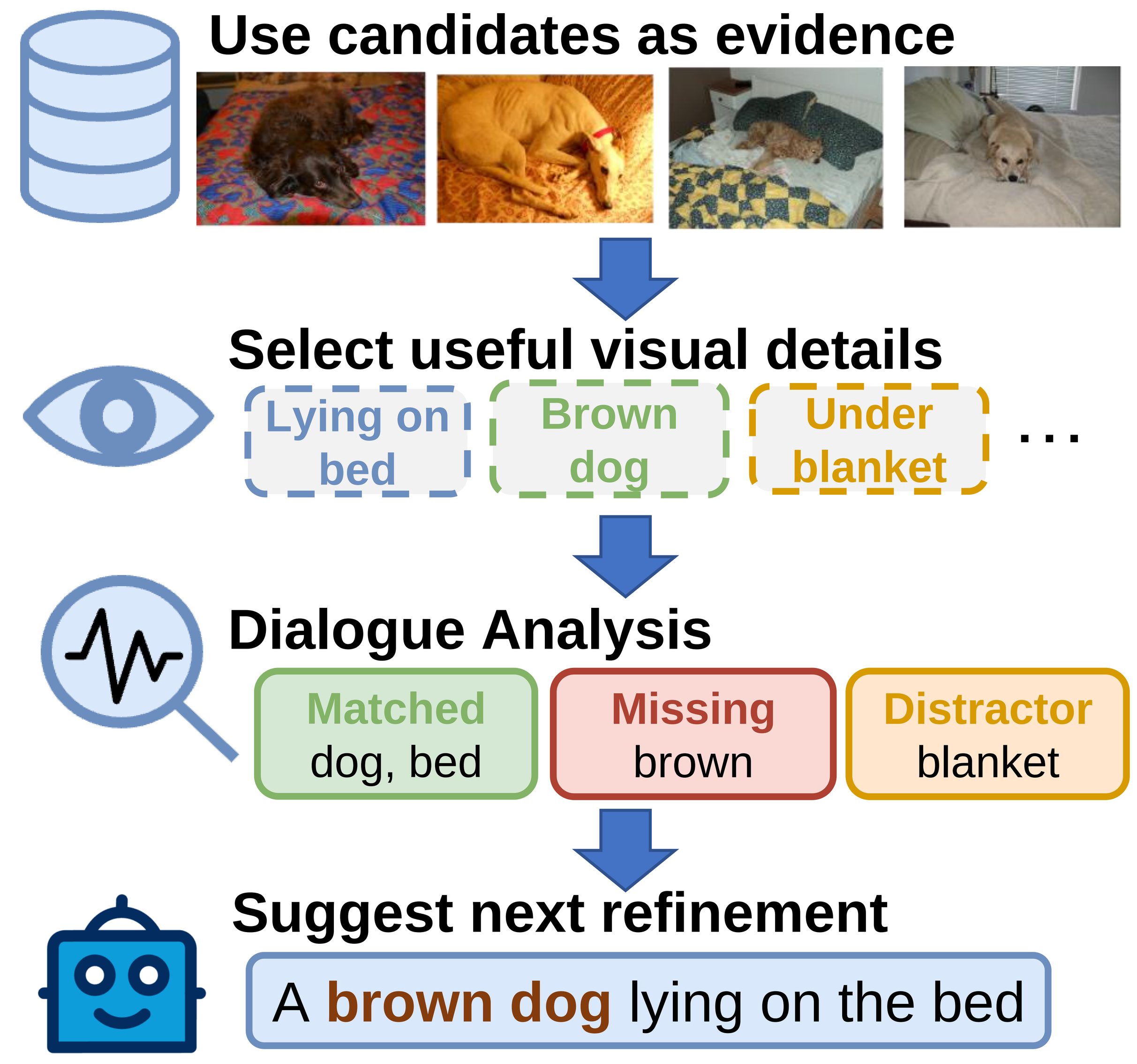
Pham Dinh Duy[1], Itthisak Phueaksri[2,1], Yasutomo Kawanishi[3,2,1], Takahiro Komamizu[1], Ichiro Ide[1]

1 Nagoya University – 2 RIKEN – 3 Ritsumeikan University

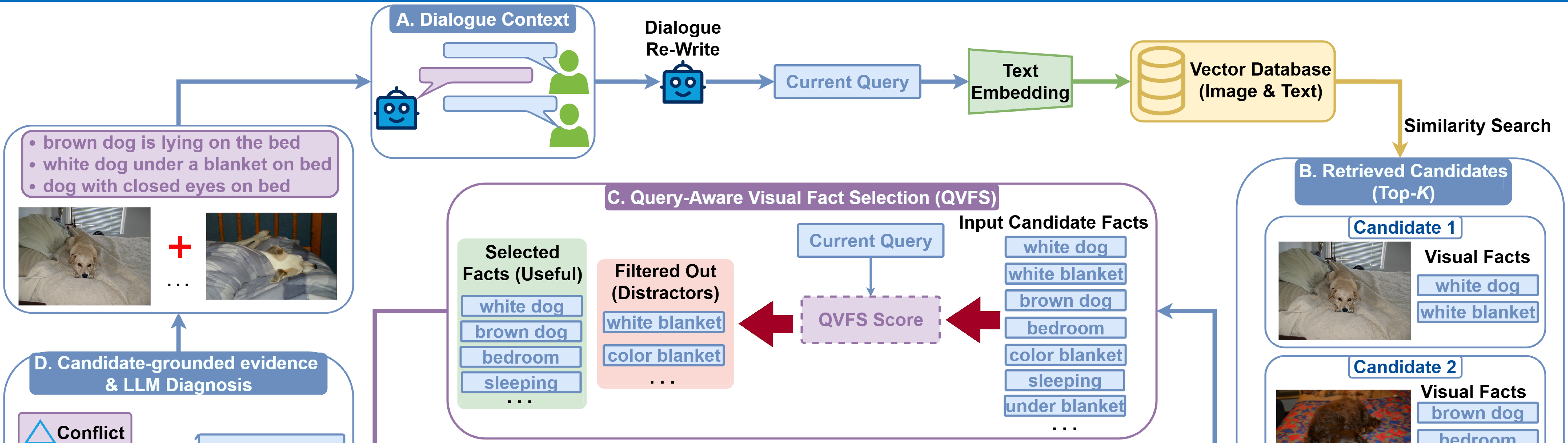
1 Traditional Retrieval

2 Goal &Technical Challenges

3 Proposed Idea



4 Proposed Framework: RAIR-VF



QVFS Score

$$\text{score}(f \mid q, \mathcal{C}) = \alpha\ \text{sim}(q, f) + \beta\ \text{overlap}(q, f) + \gamma\ \text{disc}(f, \mathcal{C}) - \delta\ \text{contra}(f, q)$$

q : Current query f : Visual fact $\mathcal{C}$: Retrieved candidate set

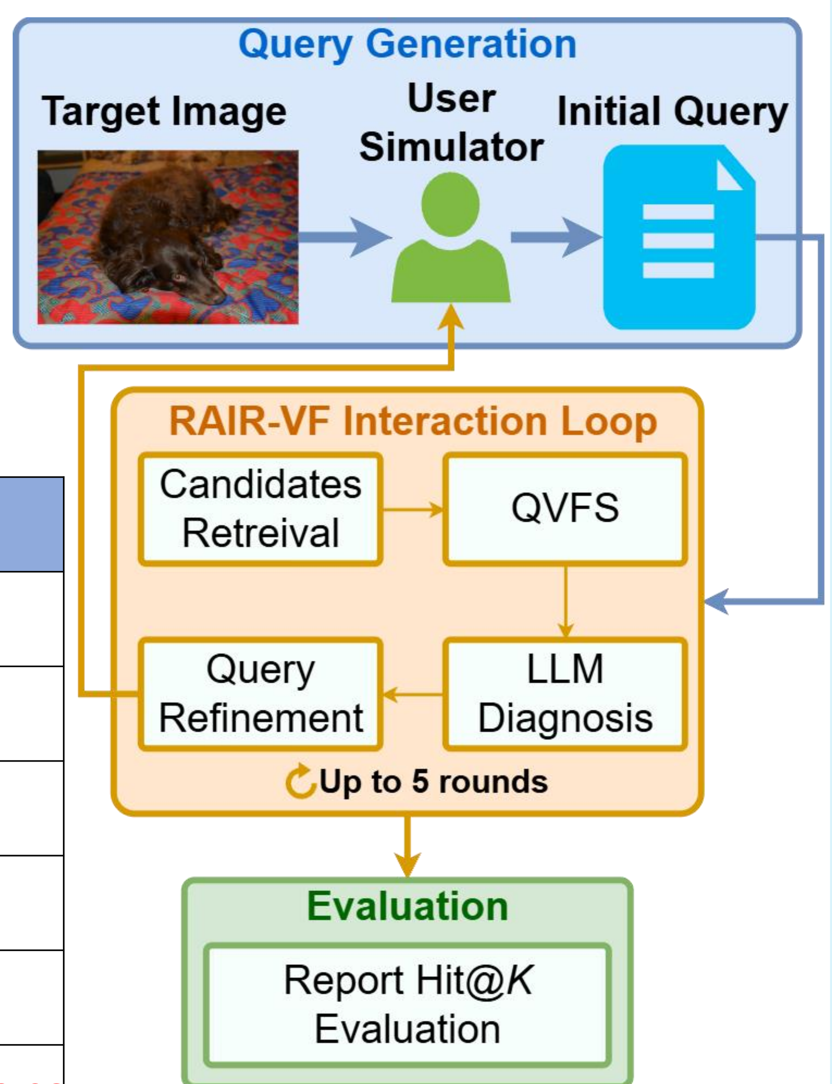
- **Semantic Similarity - sim**: Semantic relevance (text embedding similarity)
- **Lexical Overlap - overlap**: Lexical overlap with the query
- **Discriminativeness - disc**: Candidate-level discriminativeness
- **Contradiction Penalty - contra**: Contradiction penalty

RAIR-VF: Retrieval candidate images, select compact visual facts, diagnose matched / missing / conflicting constraints, and suggest refined queries.

5 Experiments & Results

Dataset: Visual Dialog (VisDial) [1]
Metric: Hit@K
Local LLM: Gemma3-12B [2]
User simulator: Qwen2.5-VL [3]
Database: PostgreSQL, CLIP-ViT [4]
Similarity Search: FAISS [5]

Turn	Hit@10	Hit@20	Hit@50
0	0.4444	0.5456	0.6769
1	0.4788	0.5794	0.7125
2	0.4913	0.6394	0.7856
3	0.5320	0.6931	0.8212
4	0.5969	0.7450	0.8531
5	0.6350 +19.06	0.7756 +23.00	0.8731 +19.62



6 Conclusion & Future Works

Conclusion

- **Candidate-grounded visual facts:** Support query refinement.
- **QVFS**: Provides compact and relevant evidence.
- **LLM diagnosis:** Helps clarify missing or conflicting constraints.
- **Multi-turn interaction**: Improves best-so-far retrieval performance.

Future Works:

- Deploying an **interactive user application** on a web application platform
- Conduct **human-user studies** to evaluate the usefulness and interpretability of refinement suggestions.

[1] A. Das, S. Kottur, K. Gupta, A. Singh, D. Yadav, J. M. F. Moura, D. Parikh, and D. Batra, "Visual Dialog," CVPR - 2017.
[2] Gemma Team, A. Kamath, J. Ferret, S. Pathak, et al., "Gemma 3 Technical Report," arXiv:2503.19786, 2025.
[3] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, et al., "Qwen2.5-VL Technical Report," arXiv:2502.13923, 2025.
[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML - 2021.
[5] J. Johnson, M. Douze, and H. Jégou, "Billion-scale Similarity Search with GPUs," IEEE Transactions on Big Data, 2019.

Contact: phamd@cs.is.i.nagoya-u.ac.jp